

Table 1: Experiment 8: SafeDreamer vs MACPO on MAMuJoCo (continuous actions).

		Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$	
2x3HalfCheetah	5.0	MACPO	Run		10M	1316.8 $\pm$ 115.7	5.0 $\pm$ 1.7	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		100k	8.5	0.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		500k	1410.6	0.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		700k	1455.9	15.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		800k	1450.3	22.2	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		900k	1685.9	37.8	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		1M	1519.2	35.8	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		100k	393.4 $\pm$ 466.5	0.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		500k	1554.6 $\pm$ 631.6	0.8 $\pm$ 1.3	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		700k	1801.6 $\pm$ 696.3	4.6 $\pm$ 4.1	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		800k	1931.8 $\pm$ 820.7	127.0 $\pm$ 208.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		900k	1744.9 $\pm$ 724.4	66.8 $\pm$ 100.4	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		1M	1480.8 $\pm$ 97.6	4.6 $\pm$ 2.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		100k	445.1 $\pm$ 454.1	0.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		500k	717.3 $\pm$ 963.0	15.4 $\pm$ 21.4	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		700k	728.8 $\pm$ 1041.0	0.6 $\pm$ 0.5	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		800k	838.8 $\pm$ 1160.6	3.2 $\pm$ 3.0	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		900k	722.3 $\pm$ 439.0	4.8 $\pm$ 7.7	
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		1M	1166.0 $\pm$ 512.6	11.1 $\pm$ 18.9	
2x3HalfCheetah	25.0	MACPO	Run		10M	1374.3 $\pm$ 85.7	22.0 $\pm$ 4.2	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		100k	1136.2 $\pm$ 564.6	30.1 $\pm$ 30.2	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		500k	2000.5 $\pm$ 120.6	5.4 $\pm$ 6.9	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		700k	2587.8 $\pm$ 181.4	22.3 $\pm$ 22.2	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		800k	2545.2 $\pm$ 212.5	37.6 $\pm$ 9.4	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		900k	2528.5 $\pm$ 162.9	3.9 $\pm$ 3.7	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		1M	2527.2 $\pm$ 228.6	16.6 $\pm$ 2.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		100k	-60.1 $\pm$ 39.0	0.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		500k	743.2 $\pm$ 408.4	0.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		700k	969.0 $\pm$ 587.9	0.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		800k	1019.7 $\pm$ 396.6	0.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		900k	1070.2 $\pm$ 308.5	0.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		1M	1149.1 $\pm$ 312.7	0.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		100k	409.9 $\pm$ 197.0	0.0	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		500k	960.4 $\pm$ 942.3	45.2 $\pm$ 52.7	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		700k	1581.8 $\pm$ 590.2	41.8 $\pm$ 40.3	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		800k	1120.2 $\pm$ 467.7	18.2 $\pm$ 17.5	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		900k	824.5 $\pm$ 464.8	2.9 $\pm$ 1.8	
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		1M	981.5 $\pm$ 1303.1	64.4 $\pm$ 103.8	
2x3HalfCheetah	25.0	MAPPO-Lag	Run		10M	1141.3 $\pm$ 35.9	0.0	
2x4Ant	0.2	MACPO	Run		10M	859.2 $\pm$ 32.7	1.2 $\pm$ 0.7	
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		100k	894.0 $\pm$ 51.4	0.0	
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		500k	1410.2 $\pm$ 394.7	2.0 $\pm$ 1.8	
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		700k	1566.0 $\pm$ 857.2	18.0 $\pm$ 11.8	
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		800k	1626.6 $\pm$ 158.9	43.5 $\pm$ 8.2	
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		900k	1902.4 $\pm$ 359.2	43.9 $\pm$ 29.7	
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		1M	2267.8 $\pm$ 134.1	24.3 $\pm$ 5.6	
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		100k	903.4 $\pm$ 27.3	0.0	
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		500k	1187.2 $\pm$ 137.5	0.0	
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		700k	1184.0 $\pm$ 499.9	5.1 $\pm$ 5.0	
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		800k	1373.5 $\pm$ 727.7	2.0 $\pm$ 2.8	
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		900k	1494.8 $\pm$ 831.1	3.7 $\pm$ 6.0	
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		1M	1618.4 $\pm$ 640.8	4.1 $\pm$ 5.8	
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		100k	887.0 $\pm$ 13.3	0.0	

Table 1 – continued

		Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$	
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		500k	815.9 $\pm$ 181.1	8.6 $\pm$ 14.8	
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		700k	826.6 $\pm$ 252.1	3.9 $\pm$ 6.8	
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		800k	1102.1 $\pm$ 223.0	2.6 $\pm$ 4.5	
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		900k	1126.3 $\pm$ 281.9	1.7 $\pm$ 3.0	
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		1M	1191.6 $\pm$ 444.6	0.9 $\pm$ 1.2	
2x4Ant	25.0	MACPO Run			10M	798.1 $\pm$ 109.4	10.5 $\pm$ 1.7	
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		100k	836.2 $\pm$ 93.2	0.0	
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		500k	1086.0 $\pm$ 412.1	6.8 $\pm$ 11.5	
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		700k	1613.5 $\pm$ 553.1	53.1 $\pm$ 47.1	
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		800k	1794.4 $\pm$ 672.1	82.9 $\pm$ 72.0	
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		900k	1838.3 $\pm$ 583.0	32.7 $\pm$ 33.7	
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		1M	1835.8 $\pm$ 678.3	17.4 $\pm$ 15.1	
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		100k	917.0 $\pm$ 58.0	0.0	
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		500k	1207.0 $\pm$ 175.0	0.1 $\pm$ 0.2	
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		700k	1156.7 $\pm$ 477.8	1.4 $\pm$ 1.2	
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		800k	1221.1 $\pm$ 478.7	2.0 $\pm$ 0.6	
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		900k	1323.1 $\pm$ 629.2	7.0 $\pm$ 7.2	
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		1M	1552.4 $\pm$ 475.2	22.6 $\pm$ 18.3	
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		100k	918.4 $\pm$ 25.2	0.0	
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		500k	1083.4 $\pm$ 114.7	0.0	
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		700k	1216.5 $\pm$ 243.5	1.0 $\pm$ 1.1	
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		800k	1221.0 $\pm$ 60.1	1.3 $\pm$ 1.3	
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		900k	1505.5 $\pm$ 345.8	4.7 $\pm$ 7.6	
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		1M	1491.0 $\pm$ 79.8	19.4 $\pm$ 30.3	
2x4Ant	25.0	MAPPO-Lag Run			10M	1579.6 $\pm$ 185.0	1.4 $\pm$ 0.3	
4x2Ant	1.0	MACPO Run			10M	1097.6 $\pm$ 97.4	3.4 $\pm$ 1.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		100k	916.3 $\pm$ 15.3	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		500k	965.2 $\pm$ 34.3	0.3 $\pm$ 0.6	
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		700k	886.0 $\pm$ 110.8	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		800k	1134.0 $\pm$ 170.2	0.1 $\pm$ 0.1	
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		900k	1140.6 $\pm$ 33.1	0.5 $\pm$ 0.8	
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		1M	1241.0 $\pm$ 565.5	2.5 $\pm$ 3.1	
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		100k	960.3 $\pm$ 33.3	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		500k	1069.5 $\pm$ 106.2	0.1 $\pm$ 0.1	
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		700k	1432.8 $\pm$ 404.7	4.2 $\pm$ 7.3	
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		800k	1378.4 $\pm$ 295.2	7.7 $\pm$ 13.4	
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		900k	1595.7 $\pm$ 410.7	2.5 $\pm$ 4.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		1M	1558.8 $\pm$ 359.0	4.2 $\pm$ 7.3	
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		100k	901.8 $\pm$ 77.0	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		500k	850.9 $\pm$ 101.8	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		700k	1105.4 $\pm$ 193.7	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		800k	1104.5 $\pm$ 140.4	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		900k	1159.3 $\pm$ 189.8	0.0	
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		1M	1032.4 $\pm$ 57.2	0.0	
4x2Ant	25.0	MACPO Run			10M	1287.3 $\pm$ 148.3	23.6 $\pm$ 6.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		100k	902.4 $\pm$ 49.0	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		500k	997.3 $\pm$ 83.4	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		700k	1192.7 $\pm$ 86.6	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		800k	1265.9 $\pm$ 101.8	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		900k	1406.8 $\pm$ 172.4	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		1M	1574.5 $\pm$ 304.2	0.5 $\pm$ 0.8	
4x2Ant	25.0	SafeDreamer	(lr=1e-4)		100k	918.7 $\pm$ 12.6	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-4)		500k	1025.5 $\pm$ 69.4	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-4)		700k	1168.3 $\pm$ 497.4	0.0	
4x2Ant	25.0	SafeDreamer	(lr=1e-4)		800k	1221.5 $\pm$ 220.3	0.1 $\pm$ 0.2	

Table 1 – continued

		Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$	
4x2Ant	25.0	SafeDreamer	(lr=1e-4)		900k	1260.2 $\pm$ 302.6		0.7 $\pm$ 1.2
4x2Ant	25.0	SafeDreamer	(lr=1e-4)		1M	1504.2 $\pm$ 520.0		2.9 $\pm$ 4.8
4x2Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		100k	939.3 $\pm$ 55.0		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		500k	1164.9 $\pm$ 175.7		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		700k	1301.5 $\pm$ 286.0		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		800k	1335.0 $\pm$ 278.6		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		900k	1515.5 $\pm$ 253.9		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		1M	1656.8 $\pm$ 161.8		0.5 $\pm$ 0.9
4x2Ant	25.0	MAPPO-Lag	Run		10M	1865.5 $\pm$ 41.9		0.1 $\pm$ 0.2